

RETAILMART V3 • SQL

Data Types, Constraints & Keys

WhatsApp Announcement

DATA TYPES + CONSTRAINTS & KEYS: Make Your Data Correct and Safe!

Yesterday every column was TEXT - great for learning structure, dangerous in real life. Today you give each column its PROPER type (and ALTER yesterday's TEXT columns into them), then add RULES that make bad data impossible.

Today's Focus:

- Data Types - INT, NUMERIC, DATE, TIMESTAMP, BOOLEAN, and the type decision framework
- ALTER TABLE ... TYPE ... USING - convert yesterday's TEXT columns to real types
- NOT NULL, DEFAULT, UNIQUE, CHECK - data integrity constraints
- PRIMARY KEY - identify every row. FOREIGN KEY - link tables. ON DELETE actions.

Important: ALL practice today is in YOUR practice database (accio_NN). You never change RetailMart's tables - you study its type choices and build your own.

Course Portal: <https://starlordali4444.github.io/postgresql/>

Today your tables go from 'a pile of text' to a real, self-defending database. Let's go!

- Siraj Sir

#Day4 #DataTypes #Constraints #Keys #PostgreSQL

TODAY'S AGENDA (2 Hours)

Section	Time	Topics
Quick Recap - Day 3 (DDL + DML)	5 mins	You built TEXT-only tables, You INSERT/UPDATE/DELETE/TRUNCATE, Why all-TEXT hurt
Topic 1: Data Types	40 mins	Numeric, Text, Date/Time, Boolean, Special, The Type Decision Framework, ALTER TEXT -> proper types with USING
Practice Block 1 - Data Types	25 mins	4 problems: pick types, fix wrong types, ALTER TEXT columns
Topic 2: Constraints & Keys	45 mins	NOT NULL, DEFAULT, UNIQUE, CHECK, PRIMARY KEY, FOREIGN KEY, ON DELETE RESTRICT / CASCADE / SET NULL
Practice Block 2 - Constraints & Keys	25 mins	4 problems: enforce rules, link tables, demo violations
Wrap-up	10 mins	Summary, self-check, homework, Filtering & Sorting preview

Where We Left Off

Yesterday you:

- Built tables with CREATE SCHEMA / CREATE TABLE (DDL)
- Gave EVERY column the type TEXT (focus on structure, not types)
- Filled tables with INSERT, changed rows with UPDATE, removed rows with DELETE, emptied tables with TRUNCATE (DML)

The pain you felt with all-TEXT columns:

- Math was clumsy: `amount * 1.1` fails (amount is TEXT)
- Comparisons sorted like words: `WHERE total > '900'` MISSED `'1000'` (`'1000' < '900'` character-by-character)

Today we fix BOTH problems:

Topic 1 (Data Types): ALTER those TEXT columns to INT/NUMERIC/DATE/BOOLEAN.

Topic 2 (Constraints & Keys): add RULES so bad data can never get in.

The Rs. 34 Lakh Bug - Why TEXT-For-Everything Is Dangerous (1/2)

A fintech startup in Bengaluru stored transaction amounts as TEXT (exactly like yesterday's tables). Their reasoning: 'We will convert to numbers when we need to calculate.' Sound familiar?

[How TEXT Destroyed Their Finances]

-- Their payment INSERTs looked fine (all TEXT):

```
INSERT INTO payments (user_id, amount) VALUES ('1001', '15000');
```

-- But one day a frontend bug sent these - and TEXT ACCEPTED them all:

```
INSERT INTO payments (user_id, amount) VALUES ('1003', 'Rs.15,000');
```

```
INSERT INTO payments (user_id, amount) VALUES ('1004', 'fifteen thousand');
```

The Rs. 34 Lakh Bug - Why TEXT-For-Everything Is Dangerous (2/2)

-- Quarter-end report:

```
SELECT SUM(amount::NUMERIC) FROM payments;
```

-- ERROR: invalid input syntax for type numeric: 'Rs.15,000'

-- The fix was one line - the RIGHT TYPE:

```
ALTER TABLE payments ALTER COLUMN amount TYPE NUMERIC(12,2)  
    USING amount::numeric;
```

-- Now 'Rs.15,000' is REJECTED at INSERT time, not discovered 8 months later.

Data types are your FIRST line of defense. The right type REJECTS bad data at the gate so you never clean it later. That is why we do NOT leave columns as TEXT in real systems - we choose the correct type for each.

Whole Numbers and Auto-IDs

Type	Range / Behavior	Use For
SMALLINT	-32,768 to 32,767	age, rating, floor number
INT/INTEGER	+/- 2.1 billion	most IDs, quantities, counts
BIGINT	+/- 9.2 quintillion	views, likes, huge counters
SERIAL	auto 1,2,3... (an auto INT)	auto-generated primary keys
BIGSERIAL	auto, BIGINT-sized	huge tables

```
-- SERIAL auto-generates ids - you never type them:  
-- INSERT INTO t (name) VALUES ('Rahul'); -- id becomes 1  
-- Deleted ids are never reused (id 2 gone forever). That is intentional.
```

Rule: use the SMALLEST integer type that comfortably fits your data.

Decimals and Money

Type	Precision	Use When
NUMERIC(p,s)	EXACT	money, prices, tax - ALWAYS
REAL / DOUBLE	approximate	scientific measurements only

NUMERIC(p, s): p = total digits, s = digits after the decimal point.

NUMERIC(10,2) -> up to 99999999.99 NUMERIC(9,6) -> GPS lat/long

-- NEVER use FLOAT/REAL for money:

SELECT 0.1::REAL + 0.2::REAL; -- 0.30000001 (WRONG)

SELECT 0.1::NUMERIC + 0.2::NUMERIC; -- 0.3 (CORRECT)

-- Over millions of transactions, float drift costs real money.

The Three Text Types

Type	Length	Best For
VARCHAR(n)	up to n	names, emails, cities, codes (most common)
TEXT	unlimited	reviews, descriptions, long comments
CHAR(n)	fixed n	almost never (pads with spaces - wasteful)

VARCHAR(100) and VARCHAR(50) use the SAME storage for a 30-char value - the limit only REJECTS values that exceed it. Set it generously.

The phone / pincode / aadhaar trap - these are NOT numbers:

phone '+91-098765-43210' -> has +, dashes, leading 0 -> VARCHAR

pincode '022987' -> INT would drop the leading 0 -> VARCHAR

Test: do you ever SUM or AVG it? No -> it is text, not a number.

Dates, Times, and Flags

Type	Stores	Use When
DATE	calendar date (2025-05-16)	birthday, order_date
TIMESTAMP	date + time	created_at, login, events
TIMESTAMPTZ	date + time + timezone	global apps
TIME	clock time only	shifts, schedules
INTERVAL	a duration (3 days 4 hours)	SLAs, time spans
BOOLEAN	true / false / NULL	is_active, has_ac flags

PostgreSQL expects dates as YYYY-MM-DD: '2025-05-16' (not 16/05/2025).

DATE vs TIMESTAMP: order_date = which DAY -> DATE.

page_view_at = exact moment -> TIMESTAMP.

Special (deep dive later): UUID (unguessable ids), JSONB (flexible nested),
ARRAY (many values in one cell). Covered on the JSON day.

Master these 8 and you handle 95% of columns:

SERIAL, INT, NUMERIC, VARCHAR, TEXT, DATE, TIMESTAMP, BOOLEAN.

The 10-Second Type Decision Tree

1. Auto-incrementing id? -> SERIAL (or BIGSERIAL)
2. Whole number you do math on? -> SMALLINT / INT / BIGINT
3. Money or precise decimal? -> NUMERIC(p,s) (never FLOAT)
4. Short text (name, email, code)? -> VARCHAR(n)
5. Long text (review, description)? -> TEXT
6. Calendar date? -> DATE
7. Exact moment? -> TIMESTAMP
8. Yes/No? -> BOOLEAN
9. Looks like a number but you
NEVER do math on it (phone/pin)? -> VARCHAR(n)

Still unsure? Default to VARCHAR, then ALTER later - which is exactly what we do next with yesterday's TEXT columns.

ALTER TABLE - Turning Yesterday's TEXT Into Proper Types (1/2)

Yesterday every column was TEXT. Today you convert them to their proper types WITHOUT dropping the table - using ALTER TABLE ... ALTER COLUMN ... TYPE ... USING. The USING clause tells PostgreSQL how to cast the existing text values.

[The Conversion Toolkit]

```
-- Convert a TEXT column to INT (cast existing values):
```

```
ALTER TABLE college.students
```

```
    ALTER COLUMN student_id TYPE INT USING student_id::int;
```

ALTER TABLE - Turning Yesterday's TEXT Into Proper Types

(2/2)

-- TEXT -> DATE:

```
ALTER TABLE college.students  
  ALTER COLUMN date_of_birth TYPE DATE USING date_of_birth::date;
```

-- TEXT -> BOOLEAN:

```
ALTER TABLE college.students  
  ALTER COLUMN is_active TYPE BOOLEAN USING is_active::boolean;
```

-- TEXT -> NUMERIC (money):

```
ALTER TABLE college.fee_payments  
  ALTER COLUMN amount TYPE NUMERIC(10,2) USING amount::numeric;
```

-- Other ALTERs you will use:

```
ALTER TABLE t ADD COLUMN created_at TIMESTAMP DEFAULT now();  
ALTER TABLE t RENAME COLUMN old_name TO new_name;  
ALTER TABLE t DROP COLUMN unused;
```

REAL WORLD

RETAILMART TYPE AUDIT - Learn From Production

How RetailMart V3 chose types (run `\d sales.orders` to see for yourself):

[sales.orders - Type Choices Explained]

Column	Type	Why
order_id	SERIAL	auto-increment primary key
cust_id	INT	references customers (a FK)
order_date	DATE	the DAY matters, not the hour
order_status	VARCHAR(30)	'Delivered', 'Processing'...
gross_total	NUMERIC(12,2)	money - MUST be exact
net_total	NUMERIC(12,2)	gross_total - discount
payment_mode_id	INT	references finance.payment_modes (FK)

Patterns: ids are SERIAL/INT, money is NUMERIC, day-only is DATE, status is VARCHAR. Every choice is deliberate - and yours should be too.

PRACTICE BLOCK 1 - Data Types

All practice in `accio_NN`. Choosing the RIGHT type is the skill - every wrong type is a future bug.

SCENARIO: Yesterday you built college.students with EVERY column as TEXT. Today, give each column its proper type - without dropping the table.

YOUR TASK:

- a)** : Make sure college.students has a few rows (from yesterday). If empty, INSERT 2-3 rows (text values like '1', '2004-03-15', 'true').
- b)** : ALTER student_id from TEXT to INT (use USING student_id::int).
- c)** : ALTER date_of_birth from TEXT to DATE.
- d)** : ALTER is_active from TEXT to BOOLEAN.
- e)** : Run \d college.students and confirm the new types. Now student_id math and date comparisons work.

Hint: TYPE ... USING column::newtype tells PostgreSQL how to convert the existing text. If a value cannot convert, the ALTER fails - that is the type depending your data.

Answer

✓ ANSWER

```
-- (a) ensure some data exists
INSERT INTO college.students
    (student_id, first_name, last_name, email, date_of_birth, is_active)
VALUES ('1', 'Aarav', 'Sharma', 'aarav@college.edu', '2004-03-15', 'true');

-- (b)(c)(d) convert TEXT -> proper types
ALTER TABLE college.students
    ALTER COLUMN student_id TYPE INT USING student_id::int;
ALTER TABLE college.students
    ALTER COLUMN date_of_birth TYPE DATE USING date_of_birth::date;
ALTER TABLE college.students
    ALTER COLUMN is_active TYPE BOOLEAN USING is_active::boolean;

-- (e) verify
SELECT column_name, data_type FROM information_schema.columns
WHERE table_schema='college' AND table_name='students'
ORDER BY ordinal_position;
```

SCENARIO: A junior developer sent this CREATE TABLE. It is full of WRONG type choices. Find and fix every mistake.

YOUR TASK: Write the corrected CREATE TABLE, execute it in accio_NN, and insert 2 rows to prove the types work.

Hint: money -> NUMERIC(12,2); phone/pincode -> VARCHAR; yes/no -> BOOLEAN; rating 1-5 -> NUMERIC(2,1); long notes -> TEXT; id -> SERIAL.

Find the 9 Wrong Types

✓ ANSWER

```
CREATE TABLE public.customer_orders (  
  order_id      VARCHAR(10),    -- ids are integers  
  customer_name CHAR(200),      -- CHAR pads; names vary  
  order_date    VARCHAR(20),    -- use a real date  
  total_amount  INT,            -- money has paisa!  
  phone         BIGINT,         -- loses +91 / leading 0  
  pincode       INT,           -- loses leading 0  
  is_delivered  VARCHAR(5),     -- true/false  
  rating        NUMERIC(10,5),  -- rating is 1-5, not 99999.99999  
  notes         VARCHAR(50)     -- notes can be long  
);
```

Answer



ANSWER

```
CREATE TABLE public.customer_orders (  
  order_id      SERIAL PRIMARY KEY, -- was VARCHAR(10)  
  customer_name VARCHAR(100),       -- was CHAR(200)  
  order_date    DATE,               -- was VARCHAR(20)  
  total_amount  NUMERIC(12,2),      -- was INT (paisa lost)  
  phone        VARCHAR(15),         -- was BIGINT  
  pincode      VARCHAR(6),          -- was INT (leading 0)  
  is_delivered  BOOLEAN,            -- was VARCHAR(5)  
  rating       NUMERIC(2,1),         -- was NUMERIC(10,5)  
  notes        TEXT                 -- was VARCHAR(50)  
);  
  
INSERT INTO public.customer_orders  
  (customer_name, order_date, total_amount, phone, pincode,  
   is_delivered, rating, notes)  
VALUES  
  ('Priya', '2026-01-05', 1299.50, '+91-9876543210', '400058',  
   true, 4.5, 'Leave at the gate.'),  
  ('Rahul', '2026-01-06', 2499.00, '9123456780', '110001',  
   false, 3.5, 'Call on arrival.');
```

SCENARIO: You are designing tables for Zomato from scratch. Millions of orders per day - every wrong type haunts you at scale. Choose types deliberately.

YOUR TASK: Create these 2 tables in `accio_NN` with correct types and justify each:

a) : restaurants - id, name, city, latitude, longitude (28.6139 needs decimals), avg_rating (4.3 possible), is_pure_veg, opening_time, created_at.

b) : menu_items - id, restaurant_id, item_name, description (can be 500 words), price, is_veg, prep_time_minutes, calories.

Hint: lat/long -> NUMERIC(9,6) (~11cm precision); rating -> NUMERIC(2,1); description -> TEXT; opening_time -> TIME; created_at -> TIMESTAMPTZ.

Answer

✓ ANSWER

```
CREATE TABLE public.restaurants (  
  restaurant_id SERIAL PRIMARY KEY,  
  name          VARCHAR(150),  
  city          VARCHAR(50),  
  latitude      NUMERIC(9,6),      -- ~11cm precision  
  longitude     NUMERIC(9,6),  
  avg_rating    NUMERIC(2,1),      -- 4.3 ok, blocks 42.3  
  is_pure_veg   BOOLEAN,  
  opening_time  TIME,  
  created_at    TIMESTAMPTZ DEFAULT now()  
);
```

```
CREATE TABLE public.menu_items (  
  item_id       SERIAL PRIMARY KEY,  
  restaurant_id INT,  
  item_name     VARCHAR(120),  
  description   TEXT,              -- up to 500 words  
  price         NUMERIC(10,2),  
  is_veg        BOOLEAN,  
  prep_time_minutes INT,  
  calories      INT  
);
```

SCENARIO: A real migration: a TEXT-everything table already has data loaded. Convert EVERY column to its proper type in place - exactly what teams do when they inherit a badly-designed table.

YOUR TASK:

a) : Create shop.legacy_orders (order_id, order_date, total, is_paid) - all TEXT - and INSERT 3 rows of clean text values.

b) : ALTER order_id TEXT->INT, order_date TEXT->DATE, total TEXT->NUMERIC(10,2), is_paid TEXT->BOOLEAN (each with USING).

c) : Now prove the fix worked: SELECT with WHERE total > 900 (numeric!) and ORDER BY order_date - both behave correctly now.

d) : Bonus: try inserting total = 'abc' AFTER the conversion - the NUMERIC type now rejects it.

Hint: This is the whole arc: yesterday's TEXT pain -> today's ALTER ... USING -> correct math and sorting forever. Keep the loaded data text-clean so every USING cast succeeds.

Answer (1/2)

✓ ANSWER

```
CREATE SCHEMA IF NOT EXISTS shop;
CREATE TABLE shop.legacy_orders (
  order_id  TEXT,
  order_date TEXT,
  total     TEXT,
  is_paid   TEXT
);
INSERT INTO shop.legacy_orders VALUES
  ('1', '2026-01-10', '900', 'true'),
  ('2', '2026-01-05', '1000', 'false'),
  ('3', '2026-01-20', '4999', 'true');
```

```
ALTER TABLE shop.legacy_orders
  ALTER COLUMN order_id  TYPE INT          USING order_id::int;
ALTER TABLE shop.legacy_orders
  ALTER COLUMN order_date TYPE DATE        USING order_date::date;
ALTER TABLE shop.legacy_orders
  ALTER COLUMN total     TYPE NUMERIC(10,2) USING total::numeric;
ALTER TABLE shop.legacy_orders
  ALTER COLUMN is_paid   TYPE BOOLEAN      USING is_paid::boolean;
```

```
-- Now numbers and dates behave correctly:
```


Answer (2/2)

✓ ANSWER

```
SELECT order_id, total FROM shop.legacy_orders
WHERE total > 900 ORDER BY order_date;
-- '1000' and '4999' both return (real numeric comparison).
```

The Exam Hall Without Rules

A school runs an exam with NO rules. Chaos:

- Two students sit in the same seat - fight (no UNIQUE)
- A student's name is blank on the admit card (no NOT NULL)
- Roll number -47 is accepted (no CHECK)
- Hall 99 does not exist but the card says it (no FOREIGN KEY)

The vice-principal fixes it: every seat a unique number (PRIMARY KEY), every student a name (NOT NULL), positive rolls only (CHECK), hall numbers must match the real list (FOREIGN KEY). Next exam: zero chaos. Those rules are CONSTRAINTS.

Application code can be bypassed. Database constraints CANNOT. A single CHECK (price > 0) can save crores.

NOT NULL & DEFAULT

NOT NULL: the column can never be empty - any INSERT/UPDATE setting NULL is rejected. DEFAULT: if you do not provide a value, PostgreSQL auto-fills one. Together they mean 'this column always has a sensible value, even if the developer forgets'.

[Syntax + Worked Example]

```
CREATE TABLE day_4.orders (  
    order_id      SERIAL PRIMARY KEY,  
    customer_name VARCHAR(100) NOT NULL,  
    order_status  VARCHAR(20)  NOT NULL DEFAULT 'pending',  
    created_at    TIMESTAMPTZ  NOT NULL DEFAULT now()  
);  
  
-- Provide only the required value; defaults fill the rest:  
INSERT INTO day_4.orders (customer_name) VALUES ('Priya');  
-- order_status='pending', created_at=now (auto-filled)  
  
-- FAILS: customer_name is NOT NULL and missing  
INSERT INTO day_4.orders (order_status) VALUES ('shipped');  
-- ERROR: null value in column "customer_name" violates not-null constraint
```

UNIQUE & CHECK (1/2)

UNIQUE: no two rows may share the value (emails, phones, PAN). Multiple NULLs ARE allowed - NULL != NULL in PostgreSQL. CHECK: a custom rule evaluated on every write; if it is false, the write is rejected.

UNIQUE & CHECK (2/2)

[Syntax + Worked Example]

```
CREATE TABLE day_4.signups (  
    signup_id SERIAL PRIMARY KEY,  
    email      VARCHAR(150) UNIQUE NOT NULL,  
    age        INT CHECK (age >= 18),  
    status     VARCHAR(20) NOT NULL DEFAULT 'active'  
    CHECK (status IN ('active','paused','closed')),  
    -- table-level CHECK can span columns:  
    start_date DATE, end_date DATE,  
    CONSTRAINT valid_dates CHECK (end_date >= start_date)  
);
```

```
INSERT INTO day_4.signups (email, age, start_date, end_date)  
VALUES ('priya@gmail.com', 25, '2026-01-01', '2026-12-31'); -- OK
```

```
-- FAILS: duplicate email (UNIQUE)  
-- FAILS: age 16 (CHECK age >= 18)  
-- FAILS: status 'banned' (CHECK IN list)
```

PRIMARY KEY - The Identity of Every Row

PRIMARY KEY = UNIQUE + NOT NULL. It uniquely identifies each row, only ONE per table. Usually a SERIAL surrogate id (immutable, compact). Use UNIQUE - not PRIMARY KEY - for natural identifiers like email (those change; a PK should never change).

[Syntax + Composite Key]

-- Single-column (most common):

```
CREATE TABLE day_4.customers (  
    customer_id SERIAL PRIMARY KEY,  
    email        VARCHAR(150) UNIQUE NOT NULL  
);
```

-- Composite PK - the COMBINATION must be unique

-- (one attendance row per employee per day):

```
CREATE TABLE day_4.attendance (  
    emp_id   INT   NOT NULL,  
    att_date DATE NOT NULL,  
    status   VARCHAR(10) NOT NULL DEFAULT 'present',  
    PRIMARY KEY (emp_id, att_date)  
);  
-- inserting the same (emp_id, att_date) twice -> duplicate key error
```

FOREIGN KEY - The Bridge Between Tables (1/2)

A FOREIGN KEY column references the PRIMARY KEY of another table. It enforces referential integrity: a child row can only reference a parent that exists - no orphan records. Insert the parent BEFORE the child.

[Syntax + Worked Example]

```
CREATE TABLE day_4.rm_customers (  
    customer_id SERIAL PRIMARY KEY,  
    full_name    VARCHAR(100) NOT NULL,  
    email        VARCHAR(150) UNIQUE NOT NULL  
);
```

FOREIGN KEY - The Bridge Between Tables (2/2)

```
CREATE TABLE day_4.rm_orders (  
  order_id      SERIAL PRIMARY KEY,  
  customer_id INT NOT NULL  
    REFERENCES day_4.rm_customers(customer_id),  
  order_total NUMERIC(10,2) NOT NULL CHECK (order_total > 0)  
);
```

```
INSERT INTO day_4.rm_customers (full_name, email)  
VALUES ('Meera', 'meera@gmail.com');  
INSERT INTO day_4.rm_orders (customer_id, order_total)  
VALUES (1, 2499.00);    -- OK, customer 1 exists
```

```
-- FAILS: customer 999 does not exist (orphan blocked)  
INSERT INTO day_4.rm_orders (customer_id, order_total) VALUES (999, 500);
```


ON DELETE Actions - What Happens to Children?

When you delete a parent row that still has children, the FK's ON DELETE clause decides the outcome:

- **RESTRICT** (default): block the delete until children are handled. Safest - use for financial/critical data.
- **CASCADE**: delete the children too. For disposable dependents (comments on a deleted post).
- **SET NULL**: keep the children, set their FK to NULL. For reassignable links (employee's manager leaves).

[Syntax]

```
FOREIGN KEY (post_id) REFERENCES day_4.blog_posts(post_id)
    ON DELETE CASCADE;
```

```
manager_id INT REFERENCES day_4.managers(manager_id)
    ON DELETE SET NULL;    -- FK column must allow NULL
```

```
dept_id INT NOT NULL REFERENCES day_4.departments(dept_id)
    ON DELETE RESTRICT;    -- the safe default
```

PRACTICE BLOCK 2 - Constraints & Keys

All practice in `accio_NN` (schema `day_4`). The failing INSERTs are intentional - read the error to see which rule fired.

SCENARIO: A food-delivery app needs every order to have a customer name, with status and timestamp auto-filled even when the app crashes mid-checkout.

YOUR TASK:

- a)** : Create day_4.orders: order_id SERIAL PK, customer_name NOT NULL, order_status DEFAULT 'pending', created_at DEFAULT now().
- b)** : INSERT a row with only customer_name; SELECT it and confirm the defaults filled in.
- c)** : Try an INSERT with no customer_name - observe the NOT NULL error.

Hint: DEFAULT applies only when you OMIT the column. NOT NULL + DEFAULT = always-present value, even if the developer forgets it.

Answer



ANSWER

```
CREATE SCHEMA IF NOT EXISTS day_4;
CREATE TABLE day_4.orders (
  order_id      SERIAL PRIMARY KEY,
  customer_name VARCHAR(100) NOT NULL,
  order_status  VARCHAR(20)  NOT NULL DEFAULT 'pending',
  created_at    TIMESTAMPTZ  NOT NULL DEFAULT now()
);

INSERT INTO day_4.orders (customer_name) VALUES ('Priya Mehta');
SELECT * FROM day_4.orders;    -- status='pending', created_at filled

-- FAILS: customer_name missing
INSERT INTO day_4.orders (order_status) VALUES ('shipped');
-- ERROR: null value in column "customer_name" violates not-null constraint
```

SCENARIO: An order-tracking system must only accept known statuses and ratings of 1-5, and never duplicate an order's tracking email.

YOUR TASK:

a) : Create day_4.order_tracking: tracking_id SERIAL PK, contact_email UNIQUE NOT NULL, status CHECK IN ('placed','shipped','delivered','cancelled'), rating CHECK BETWEEN 1 AND 5.

b) : INSERT one valid row.

c) : Show three failing INSERTs: a duplicate email, status 'dispatched', and rating 7.

Hint: Each constraint is checked independently. CHECK (status IN (...)) is a dropdown; CHECK (rating BETWEEN 1 AND 5) is a range.

Answer



```
CREATE TABLE day_4.order_tracking (  
    tracking_id    SERIAL PRIMARY KEY,  
    contact_email  VARCHAR(150) UNIQUE NOT NULL,  
    status         VARCHAR(20) NOT NULL  
        CHECK (status IN ('placed','shipped','delivered','cancelled')),  
    rating         INT CHECK (rating BETWEEN 1 AND 5)  
);  
  
INSERT INTO day_4.order_tracking (contact_email, status, rating)  
VALUES ('a@shop.com', 'shipped', 4);    -- OK  
  
-- FAILS (UNIQUE): duplicate email  
INSERT INTO day_4.order_tracking (contact_email, status) VALUES ('a@shop.com','placed');  
-- FAILS (CHECK): 'dispatched' not allowed  
INSERT INTO day_4.order_tracking (contact_email, status) VALUES ('b@shop.com','dispatched');  
-- FAILS (CHECK): rating 7 out of range  
INSERT INTO day_4.order_tracking (contact_email, status, rating) VALUES ('c@shop.com','placed',7);
```

SCENARIO: RetailMart needs orders that always belong to a real customer - no ghost orders pointing at non-existent people.

YOUR TASK:

a) : Create day_4.rm_customers (customer_id SERIAL PK, full_name NOT NULL, email UNIQUE NOT NULL).

b) : Create day_4.rm_orders with customer_id INT NOT NULL REFERENCES rm_customers, and order_total NUMERIC CHECK > 0.

c) : Insert a customer, then a valid order for them.

d) : Try an order for customer_id = 999 - observe the FK error. Then try a duplicate customer email.

Hint: Parent before child: insert the customer first. The FK type (INT) must match the parent PK (SERIAL = INT).

Answer



```
CREATE TABLE day_4.rm_customers (  
    customer_id SERIAL PRIMARY KEY,  
    full_name    VARCHAR(100) NOT NULL,  
    email        VARCHAR(150) UNIQUE NOT NULL  
);  
CREATE TABLE day_4.rm_orders (  
    order_id     SERIAL PRIMARY KEY,  
    customer_id INT NOT NULL REFERENCES day_4.rm_customers(customer_id),  
    order_total  NUMERIC(10,2) NOT NULL CHECK (order_total > 0)  
);  
  
INSERT INTO day_4.rm_customers (full_name, email)  
VALUES ('Meera Reddy', 'meera@gmail.com');  
INSERT INTO day_4.rm_orders (customer_id, order_total) VALUES (1, 2499.00); -- OK  
  
-- FAILS (FK): customer 999 does not exist  
INSERT INTO day_4.rm_orders (customer_id, order_total) VALUES (999, 500.00);  
-- FAILS (UNIQUE): duplicate email  
INSERT INTO day_4.rm_customers (full_name, email) VALUES ('X','meera@gmail.com');
```


SCENARIO: A blogging platform: deleting a post should auto-remove its comments (they are meaningless alone), but deleting a department must be BLOCKED while employees still belong to it. Pick the right ON DELETE policy for each.

YOUR TASK:

a) : Create day_4.blog_posts + day_4.post_comments with ON DELETE CASCADE. Insert a post with 3 comments, delete the post, prove comments are gone (count = 0).

b) : Create day_4.departments + day_4.dept_employees with ON DELETE RESTRICT. Insert a department + an employee, then try to delete the department - observe it is blocked.

c) : Explain in one line why CASCADE fits comments but RESTRICT fits departments (and why financial records should use RESTRICT).

Hint: Ask: if the parent disappears, does the child still make sense alone? No -> CASCADE. Yes / must be preserved -> RESTRICT.

Answer (1/2)

✓ ANSWER

```
-- (a) CASCADE: comments die with the post
CREATE TABLE day_4.blog_posts (
  post_id SERIAL PRIMARY KEY, title VARCHAR(200) NOT NULL);
CREATE TABLE day_4.post_comments (
  comment_id SERIAL PRIMARY KEY,
  post_id INT NOT NULL REFERENCES day_4.blog_posts(post_id) ON DELETE CASCADE,
  comment_text TEXT NOT NULL);
INSERT INTO day_4.blog_posts (title) VALUES ('Learn SQL');
INSERT INTO day_4.post_comments (post_id, comment_text)
VALUES (1,'Great!'),(1,'Helpful!'),(1,'Bookmarked. ');
DELETE FROM day_4.blog_posts WHERE post_id = 1;
SELECT count(*) FROM day_4.post_comments; -- 0 (auto-deleted)

-- (b) RESTRICT: cannot delete a department with employees
CREATE TABLE day_4.departments (
  dept_id SERIAL PRIMARY KEY, dept_name VARCHAR(50) UNIQUE NOT NULL);
CREATE TABLE day_4.dept_employees (
  emp_id SERIAL PRIMARY KEY, emp_name VARCHAR(100) NOT NULL,
  dept_id INT NOT NULL REFERENCES day_4.departments(dept_id) ON DELETE RESTRICT);
INSERT INTO day_4.departments (dept_name) VALUES ('Engineering');
INSERT INTO day_4.dept_employees (emp_name, dept_id) VALUES ('Arun', 1);
-- FAILS (RESTRICT): employees still reference dept 1
```

Answer (2/2)

✓ ANSWER

```
DELETE FROM day_4.departments WHERE dept_id = 1;
```

Key Takeaways (1/2)

- 1. The right type is defense:** NUMERIC for money (never FLOAT), VARCHAR for phone/pin (never INT), DATE/TIMESTAMP for time, BOOLEAN for yes/no, SERIAL for auto-ids.
- 2. ALTER ... TYPE ... USING:** Convert yesterday's TEXT columns to proper types in place. A failing cast forces you to clean bad data - the type defending itself.
- 3. NOT NULL & DEFAULT:** NOT NULL forbids empty; DEFAULT auto-fills. Together = always a sensible value.

Key Takeaways (2/2)

- 4. **UNIQUE & CHECK:** UNIQUE blocks duplicates (multiple NULLs allowed). CHECK enforces custom rules; pair with NOT NULL since NULL passes a CHECK.
- 5. **PRIMARY KEY:** UNIQUE + NOT NULL, one per table. Prefer a SERIAL surrogate; use UNIQUE for natural ids like email.
- 6. **FOREIGN KEY:** Child references an existing parent - no orphans. Parent first, child second. FK type must match parent PK type.
- 7. **ON DELETE:** RESTRICT (block, safest), CASCADE (delete children), SET NULL (unlink). Financial data -> RESTRICT + soft-delete.

Data Types, Constraints & Keys - Quick Reference

-- Core Data Types

SERIAL	auto-id (1,2,3...)
INT / BIGINT	whole numbers / math
NUMERIC(p,s)	money / exact decimals
VARCHAR(n)	short text; TEXT = long text
DATE	calendar day
TIMESTAMP(TZ)	exact moment
BOOLEAN	true / false

-- ALTER TEXT -> Proper Type

```
ALTER TABLE t ALTER COLUMN c
  TYPE INT USING c::int;
ALTER TABLE t ALTER COLUMN c
  TYPE DATE USING c::date;
ALTER TABLE t ALTER COLUMN c
  TYPE NUMERIC(10,2) USING c::numeric;
```

-- Column Constraints

```
col TYPE NOT NULL
col TYPE DEFAULT value
col TYPE UNIQUE
col TYPE CHECK (col > 0)
CHECK (status IN ('a','b'))
CONSTRAINT name CHECK (end >= start)
```

Data Types, Constraints & Keys - Quick Reference

-- Keys

```
id SERIAL PRIMARY KEY
PRIMARY KEY (col_a, col_b) -- composite
fk INT REFERENCES parent(id)
FOREIGN KEY (fk) REFERENCES parent(id)
email VARCHAR(150) UNIQUE -- natural id
```

-- ON DELETE Actions

```
... REFERENCES p(id) ON DELETE RESTRICT
... REFERENCES p(id) ON DELETE CASCADE
... REFERENCES p(id) ON DELETE SET NULL
RESTRICT = block (safe default)
CASCADE = delete children
SET NULL = keep children, unlink
```

-- Golden Rules

```
Money -> NUMERIC, never FLOAT
Phone/pin -> VARCHAR, never INT
Every table needs a PRIMARY KEY
FK type must match parent PK type
Pair CHECK with NOT NULL
Financial FKs -> RESTRICT + soft-delete
```

INTERVIEW QUESTION BANK - Data Types, Constraints & Keys

Practice answering OUT LOUD. Interviews test how clearly you explain.

Interview Questions

Q1: 'Why never store money as FLOAT?'

A: 'FLOAT/REAL are binary approximations - $0.1+0.2$ becomes 0.30000001 . Over millions of transactions the drift becomes real losses. NUMERIC stores exact decimals, so money math is always correct.'

Q2: 'Why store phone numbers as VARCHAR, not INT?'

A: 'You never do math on a phone number. It can contain +, dashes, and leading zeros that INT would strip - 098765 becomes 98765. Any identifier you never compute on should be text.'

Q3: 'How do you change a TEXT column to NUMERIC when data exists?'

A: 'ALTER TABLE t ALTER COLUMN c TYPE NUMERIC USING c::numeric. The USING clause casts existing values. If a value cannot cast, the ALTER fails - which forces me to clean the bad data first.'

Q4: 'Difference between UNIQUE and PRIMARY KEY?'

A: 'A table can have many UNIQUE constraints but only one PRIMARY KEY. UNIQUE allows NULLs; PRIMARY KEY is UNIQUE + NOT NULL. PK is the row's definitive identity; UNIQUE is business-level uniqueness like email.'

Q5: 'Does CHECK (rating BETWEEN 1 AND 5) block NULL?'

Data Types, Constraints & Keys Self-Check (12 Questions)

Close your notes. Answer from memory:

- 1. Which type for money, and which must you never use?
- 2. Why is phone VARCHAR, not INT? Give 2 reasons.
- 3. Write the ALTER that converts a TEXT column to DATE.
- 4. What does the USING clause do in ALTER ... TYPE?
- 5. Difference between NOT NULL and DEFAULT?
- 6. Does a CHECK constraint block NULL? How do you also block NULL?
- 7. UNIQUE vs PRIMARY KEY - two differences.
- 8. What is PRIMARY KEY equal to (in terms of other constraints)?
- 9. What does a FOREIGN KEY guarantee?
- 10. Parent or child first when inserting? Why?
- 11. RESTRICT vs CASCADE vs SET NULL - one use case each.
- 12. Why use RESTRICT + soft-delete for financial data?

Score 10+ -> ready for Filtering & Sorting. Below 10 -> re-read the weak sections.

Data Types, Constraints & Keys Take-Home Challenge

Complete these BEFORE Filtering & Sorting, in accio_NN:

- 1. Convert:** : Take the TEXT tables you built on Day 3 and ALTER each column to its proper type (USING casts). Confirm with information_schema.columns that none are 'text' anymore.
- 2. Constrain:** : Add NOT NULL, DEFAULT, UNIQUE, and at least one CHECK to your products/customers/orders tables.
- 3. Keys:** : Give every table a PRIMARY KEY. Add a FOREIGN KEY from orders to customers.
- 4. Prove it:** : Write 4 INSERTs that each fail for a DIFFERENT reason (NOT NULL, UNIQUE, CHECK, FK). Save the error messages.
- 5. Cascade:** : Add an order_items table with ON DELETE CASCADE to orders. Delete an order; confirm its items vanish.

Tomorrow: Filtering & Sorting - your FIRST real DQL. WHERE, LIKE, BETWEEN, IN, IS NULL, ORDER BY, LIMIT on RetailMart.

YOUR SQL JOURNEY



SQL Intro



Installation



DDL + DML



Data Types + Constraints <- YOU ARE HERE ~~RE HERE~~



Filtering (first DQL!)



Joins



CTEs



Window Functions



Indexing



Views



Cohort/RFM



Capstone

Coming Tomorrow

You have built tables, given them proper types, and protected them with constraints and keys. But how do you FIND a specific customer? List orders above Rs. 5,000? See the top 10 products by price?

Filtering & Sorting introduces WHERE, LIKE, BETWEEN, IN, IS NULL, ORDER BY, and LIMIT - run against the REAL RetailMart data (accio_retailmart_NN). This is where SQL starts to feel POWERFUL - and where you stop building and start ASKING QUESTIONS.

- See you on Filtering & Sorting! Siraj Sir